INCH-POUND

MIL-PRF-40039E
2 September 1997
SUPERSEDING
MIL-C-40039D
20 October 1981

PERFORMANCE SPECIFICATION

CLOTH, WATERPROOF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of waterproof cloth.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 4 and 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 4 and 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-595	- Colors Used In Government Procurement

MILITARY

MIL-STD-1487	- Glossary of Cloth Coating Imperfections

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Defense Personnel Support Center, Clothing and Textiles Directorate, Attn: DPSC-FNS, 2800 South 20th Street, Philadelphia, PA 19145-5099.

AMSC N/A **FSC 8305**

DISTRIBUTION STATEMENT A. ***Approved for public release; distribution is unlimited.***

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(Unless otherwise indicated. copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk. Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

FEDERAL TRADE COMMISSION

Rules and Regulations Under-the Textile Fiber Products Identification Act

(Copies are available from the Federal Trade Commission, Pennsylvania Avenue at Sixth Street, N.W., Washington, DC 20580-0001.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

(Applications for copies of referenced documents should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19426-2959.)

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

(Applications for copies of referenced documents should be addressed to TAPPI Press, Technology Park/Atlanta, P.O. Box 105113, Atlanta, GA 30348-5113.)

SEARS ROEBUCK AND COMPANY FABRIC DEFECT SCALES

(Fabric defect replica kits are available from Sears Roebuck and Company, Department 817, (ATTN: BSC 23-29), Sears Tower, Chicago, IL 60684).

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Standard sample. The waterproof cloth shall match the standard sample for shade and appearance and shall be equal to or better than the sample for colorfastness (see 6.4).

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3.3 Performance requirements. The cloth shall conform to the requirements specified in Table I and 3.4 through 3.9.

TABLE I. Requirements

Characteristic	Requirements
Weight Oz/yd^2 (max)	7.3
Breaking Strength, lbs. (min):	
Warp	90
Filling	80
Tearing Strength, grams (min):	
Warp	770
Filling	575
Hydrostatic Resistance psi (min):	
Initial	40
After strength of coating	20
After water immersion	20
Adhesion of coating, lbs/2 inch width (min) 1/	
Initial	6.0
After water immersion	5.0
Stiffness, cm (warp only) (max):	8.0
Blocking, scale rating (max)	No. 2
Resistance to: Leakage (min.)	No leakage

1/ Requirement applies if a film or coating is applied to the surface of the cloth.

3.4 Abrasion resistance. The cloth shall show no loose fibers of the base cloth exposed in the center one (1) inch of the abraded area.

3.5 Low temperature resistance. The cloth shall show no indication of the finish (if used) to crack, flake or separate from the base cloth.

3.6 Accelerated weathering resistance. The cloth shall show no increase in softness, stiffness, tackiness or brittleness when compared with an unexposed specimen of the finished cloth.

3.7 Color. The color of the cloth shall be Camouflage Green 483 conforming to color chip 34094 of FED-STD-595 (see 3.2 and 6.3).

3.7.1 Color matching. The color of the cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 $\pm$ 200^{o}K, with illumination of 100 $\pm$ 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 $\pm$ 200^{o}K.

3.8 Lumps and knots. The cloth shall be free of solid lumps, defined as a slub C or 4, or knots which exceed level C on the respective Sears Fabric Defect Scale (see 2.3 and 4.3.1.1).

3.9 Mildew resistance. Plasticizers, if used, shall be restricted to phosphate and phthalate ester plasticizers only for the purpose of mildew resistance (see 4.4.11).

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3.10 Toxicity. The cloth shall not be toxic to the skin, eyes or epidermis when used as intended (see 4.4.10).

3.11 Face side identification. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" on each end of each individual piece.

3.12 Fiber identification. Each roll shall be labeled and ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.13 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When a first article is required (see 6.2) it shall be examined for the defects listed in Table II and shall be tested for the characteristics specified in Table III.

4.3 Conformance inspection. Sampling for inspection shall be as specified in the contract or purchase order.

4.3.1 End item examination.

4.3.1.1 Yard by yard examination. During the yard by yard examination, each roll in the sample shall be examined for the defects listed below. The yardage of each roll in the sample shall be examined on both sides. The yardage shall be given a through-light inspection for pinholes and thin areas. The through-light inspection shall be performed in accordance with MIL-STD-1487. The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only one defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The sample unit shall be one linear yard.

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Table II. End item visual defects

Defect	Classification	
	Major	Minor
Any hole, cut, tear, scratch	101	
Any pinhole	102	
Any unfinished area (if finished)	103	
Any thinly finished area (if finished)	104	
Abrasion resulting in a thin or weak place	105	
Multiple floats or skips, 1/2 inch or more in either warp or filling direction	106	
Blisters, tunnels, bubbles or separation of finish (if finished)	107	
Crease or wrinkle resulting in doubling or adhesion of surfaces that cannot be corrected by manual pressure, adhesion of surfaces against each other, or any diagonal distortion of face side surface	108	
Any solid lump, defined as a slub C or 4, or knot which exceeds level C on the respective Sears Fabric Defect Scale (see 3.8)	109	
Fabric edges rolled, folded, doubled, scalloped or wavy	110	
Any spot, stain or foreign matter 1/		201
Any odor other than that which is characteristic of the component materials of the cloth		202
Any color off shade, not uniform, mottled or spotted		203
Any nonuniformity of color (color not showing solid coverage)	111	
Any tackiness	112	
Any scorch or burn	113	
Not clean	114	

1/ Clearly visible at normal inspection distance (approximately 3 feet).

4.3.1.2 Roll identification examination. During the yard-by-yard examination, each roll in the sample shall be examined for proper identification. Any roll in the sample that is not labeled or ticketed in accordance with the Textile Fiber Products Identification Act or any piece not marked with the word "Face", when required, shall be considered a roll identification defect.

4.3.1.3 Shade and appearance examination. During the yard-by-yard examination, each roll in the sample shall be examined on the face side for shade and appearance. If any roll in the sample is off shade or shaded side to side, side to center, or end to end, or if any roll does not have the same appearance as the standard sample, it shall be considered a shade and appearance defect.

4.3.2 End item testing. The waterproof cloth shall be tested for the characteristics listed in Table III. The methods of testing specified wherever applicable and as listed in Table III shall be followed. The sample unit for testing shall be 3 continuous yards full width of the finished cloth. The lot shall be unacceptable if any sample unit fails to meet any requirement specified. All test reports shall contain the individual values utilized in expressing the final results.

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE III. End item tests

Characteristic	Test method
Weight	ASTM D 3776 Method C (small swatch of fabric method)
Breaking strength	ASTM D 5034 (G-E or G-T)
Tearing strength	ASTM D 1424
Hydrostatic Resistance, Initial After strength of coating After water immersion	4.4.1 5972 1/ & 4.4.1 4.4.2
Adhesion of coating, Initial After water immersion	4.4.3.1 4.4.3.2
Stiffness, warp only:	4.4.4
Blocking	4.4.5
Resistance to leakage	4.4.6
Abrasion resistance	4.4.7
Low temperature resistance	4.4.8
Accelerated weathering resistance	4.4.9
Toxicity	4.4.10
Mildew resistance	4.4.11

1/ Except that the specimen shall be stretched at 20 pounds.

4.4 Methods of inspection.

4.4.1 Hydrostatic resistance. ASTM D-751, Hydrostatic Resistance Procedure A (Pressure Application by Mullen Type Hydrostatic Tester) Procedure 1 using a machine with displacement rate of 85ml/min. The water pressure shall be applied to the face side of the cloth.

4.4.2 Water immersion procedure. The test specimen shall be immersed in distilled water at 70°F ± 2°F for 2 hours, then excess water shall be lightly blotted from the surface and the specimen immediately tested for hydrostatic resistance in accordance with 4.4.1.

4.4.3 Adhesion of coating.

4.4.3.1 Initial. ASTM-D-751, Adhesion of Coating with 2 inch wide reinforced coating adhesion specimens, cyanoacrylate (solventless) adhesive, and pulling clamp speed of 5 mm/s. Three specimens shall be tested by adhering face-to face.

4.4.3.2 After water immersion. The test specimens of 4.4.3.1, after being tested for dry adhesion with separation of 3 inches, shall be immersed in distilled water at 70°F ± 2°F for 2 hours, removed and blotted dry. The adhesion after wet immersion shall then be immediately determined by separation of the remainder of the test specimen in accordance with 4.4.3.1.

4.4.4 Stiffness. TAPPI Method T-451, Preferred Procedure (1). except that five test specimens with the long dimension parallel to the warp direction of the finished cloth shall be tested and that the standard textile test conditions as specified in ASTM D-1776 shall be used.

4.4.5 Blocking. ASTM D-751, Determination of Blocking Resistance of Fabrics Coated with Rubber or Plastics at Elevated Temperatures, except that only one specimen shall be exposed at an oven temperature of 180°F ±2°F for 30 minutes.

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4.4.6 Resistance to leakage. ASTM-D-751, Hydrostatic Resistance, Procedure B, (Pressure Application by a Rising Column of Water), Procedure 2 with the hydrostatic head fixed at 50 centimeters and applied to the test specimen for 10 minutes. The face side of the finished cloth shall contact the water. The report shall only include "measurement of the appearance of water drops". Leakage of any specimen shall be considered a test failure. Leakage is defined as the appearance of water at three or more different places within the 4 1/2 inch diameter test area.

4.4.7 Abrasion resistance . ASTM-D-4157. The back side of the waterproof cloth shall be abraded for ten (10) cycles under a tension of six (6) pounds and pressure on the abrasion area of three (3) pounds. The abradant shall be "fine" emery cloth. Only three (3) specimens in the warpwise direction shall be tested and then visually examined.

4.4.8 Low temperature resistance. Method 5874 of FED-STD-191 except that the test for water resistance shall not be performed and that the test specimens shall be conditioned for 35 $\pm$ 5 minutes at the low temperature of 0°F $\pm$ 5°F prior to testing. The test shall then be conducted at the low temperature and the specimens visually examined.

4.4.9 Accelerated weathering resistance. Method 5804 of FED-STD-191. The face side of the finished cloth shall be exposed to the light source of the weatherometer for 200 hours. After the exposure period, the finished cloth shall be evaluated for conformance to the requirement.

4.4.10 Toxicity assessment. The waterproof cloth shall be tested for toxicity as follows:

a) Title 40, Code of Federal Regulations, 1994 Edition;
   Section 798.4100- Dermal Sensitization
   Section 798.4470- Primary Dermal Irritation
   Section 798.4500- Primary Eye Irritation
   Marzulli, F. & H. Maibach, "Contact Allergy:
   Predictive Testing in Humans", Advances in Modern Toxicology, Volume 4, pp 353-372, 1977.
b) As an alternative to animal and human testing, the contractor may provide information which certifies that the finished product was composed of chemicals or materials which have been safely used commercially where prolonged skin contact has occurred.

4.4.11 Mildew resistance. A certificate of compliance shall be submitted for this requirement.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The waterproof cloth is intended for use in the fabrication of heavy duty wet weather ponchos.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When first article is required (see 3.1 and 4.2).
d. Width required.
e. Levels of preservation and packing (see 5.1).

6.3 Standard sample. For access to the shade sample (see 3.2) address the contracting activity issuing the invitation for bids.

6.4 Subject term (key word) listing.

Water resistant
Poncho
Wet weather clothing system

Custodians:
Army - GL
Navy - NU

Review activities:
Navy - MC

Preparing activity
DLA-CT

(Project 8305-0626)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-40039D	**2. DOCUMENT DATE** ***(YYMMDD)*** 97 September 02

3. DOCUMENT TITLE

CLOTH, WATERPROOF

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME DEFENSE PERSONNEL SUPPORT CENTER DPSC-FNS	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON (215) 737-8079 444-8079
c. ADDRESS *(Include Zip Code)* 2800 South 20th STREET PHILADELPHIA, PA 19145-5099	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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